

Single-Key Cut-and-Choose for Keyed-Verification Blind-Signature Credentials: A Compact Construction and its Security Intuition

Abstract

A blind signature lets a user obtain a credential without revealing its message to the signer. In the BLNS23 scheme, proving that the user’s request is well formed is expensive because the proof must check hash computations. Replacing this proof with cut-and-choose run in parallel lanes avoids it entirely. This document explores a compact form of that idea: the signer holds *one* key, and the user commits to all branches at once by summing them into a single aggregate, binding each branch individually by a hash. The user sends the aggregate, the seeds of the branches selected for checking, and the hashes of the unopened branches; the challenge is taken over the aggregate and the full list of branch hashes, so the branch table is fixed before the challenge exists. The signer *peels* the opened branches off the aggregate and signs the remaining sum. The first message then carries one commitment vector, κ seeds, and κ hashes: with the BLNS23 keyed-verification parameters and $\kappa = 512$ lanes, about 143 KiB. We give the construction as a tuple of algorithms, its correctness calculation, drawings of the exchange, and a game-based random-oracle analysis of its security core. The design’s key property is *structured-sum fixpoint hardness*: roughly, that no attacker can force the signed sum into a chosen shape without about $2^{\kappa/2}$ work. We prove this property for the challenge game in the classical random-oracle model (Theorem 5.4): an attacker making Q_h challenge-hash queries and completing Q sessions aims a peeled sum at a committed target set T , or places two accepted sessions in a committed relation T_{rel} , only with probability $O(Q_h(|T| + Q|T_{\text{rel}}|)2^{-\kappa})$, and every accepted session retains more than $\kappa/2$ honestly formed, challenge-selected branch vectors except with probability $Q_h 2^{-\kappa/2}$. The remaining lattice step is isolated as a single decisional assumption, *structured-sum HMLWE* (Game 4): key recovery reduces to it, linear distinguishing strategies reduce to module-SIS or to the proven bounds, residual control over the samples is priced by density, one-more unforgeability follows from it together with the common-message binding, and we explain why no reduction to plain MLWE can reach it. Blindness against a dishonest signer is carried through in the same style (Section 5.9): every hop is proven in the random-oracle model except one, isolated as *short-decomposition indistinguishability* (SDI) — deciding whether the peeled sum, minus the presented lane images, has a short decomposition under B — and SDI is also necessary, so blindness is sandwiched between SDI at width 1 and at width κ (Theorem 5.13). SDI is a normal-form decision-MLWE variant, but its best known attack, a dual lattice distinguisher, sits below the lane floor at the current parameters, so blindness rather than splicing may be the binding constraint. The ssHMLWE assumption itself, the quantum analogues of both assumptions, and the simulation details of the one-more hybrids remain open (Section 7). Under that assumption the best attacks we know are the copying-and-combining attacks of Section 5.7, costing about $2^{\kappa/2}$ classical or $2^{\kappa/4}$ quantum work, so $\kappa = 512$ is the starting point.

Security status. The challenge-game core of the construction is proven in the classical random-oracle model (Section 5.5, Theorem 5.4). The remaining lattice step is isolated as one decisional assumption, structured-sum HMLWE, with the surrounding links proven: key recovery and one-more unforgeability reduce to it, and no reduction from a standard assumption is available along known lines (Section 5.6). Blindness is proven up to one further isolated assumption, short-decomposition indistinguishability, a normal-form decision-MLWE variant whose concrete costing at the current parameters may bind below the lane floor (Section 5.9). The proof does not extend to quantum query access, the isolated assumptions are unproven, and the one-more simulation details are uninstantiated (Section 7). The scheme must not be deployed as a secure credential system.

1 Introduction

In the keyed-verification blind signature of Beullens, Lyubashevsky, Nguyen, and Seiler (BLNS23) [1], the user’s first message contains a zero-knowledge proof that a lattice commitment was formed correctly, including correct evaluations of hash functions. That proof dominates the cost of issuance.

Cut-and-choose offers a different route: the user prepares many candidate *branches*, reveals some of them for checking, and the signer uses only the unchecked rest. Run in κ parallel *lanes*, in the spirit of parallel-instance cut-and-choose [2], the idea replaces the well-formedness proof with per-lane spot checks. This document explores a compact form of the construction:

- **One key.** The signer holds a single secret vector $\mathbf{s}$ and a single public matrix B . There are no lane keys.
- **One aggregate commitment, bound leaf by leaf.** The user sums all 2κ branch vectors into a single aggregate $\mathbf{c}_{\text{agg}}$ and commits to each branch individually by a *leaf hash*. The cut-and-choose challenge is derived from the aggregate together with all 2κ leaf hashes, so the branch table is committed before the challenge exists. The individual branch vectors are never transmitted: opened branches are regenerated from seeds, unopened branches are committed by their hashes only.
- **Peeling.** The signer recomputes the challenge, expands the opened branches from their seeds, subtracts them from the aggregate, and signs the remaining sum with its one key.

The appeal is concrete. With the BLNS23 keyed-verification parameters and $\kappa = 512$ lanes, the first message carries a single commitment vector, κ seeds, and κ leaf hashes: about 143 KiB, none of it a proof of hash evaluations. The signer’s response proof covers a single linear key relation, and the signer stores one key.

The cost is equally concrete, and we state it up front: the BLNS23 proof route — extraction from the user’s well-formedness proof — no longer exists, because cut-and-choose removed that proof, and the selected branches are never revealed in any extractable form: their leaf hashes commit without opening. No replacement reduction is known, and Section 5.6 explains why none can exist along known lines. Security here rests on the interaction of the challenge hash with the aggregate, which we call *structured-sum fixpoint hardness*; we prove its challenge-game content in the classical random-oracle model (Section 5.5), and we isolate the step from its conclusion to lattice hardness as a single decisional assumption, proving the links around it (Section 5.6).

Roadmap. Section 2 fixes notation, including the game-playing conventions used throughout. Section 3 gives the construction as a tuple of algorithms, with drawings. Section 4 shows why honest credentials verify. Section 5 gives the security analysis: the commit-before-challenge order, the fixpoint wall, the sum-to-target (Wagner) analysis, a random-oracle proof of the fixpoint game, the isolation of the remaining lattice step, the splicing attacks and their costs, the common-message strengthening that demotes splicing to duplication, and a blindness proof that isolates its remaining content as a second decisional assumption (Section 5.9). Section 6 estimates sizes. Section 7 lists what remains unproven.

2 Notation and basic tools

We work in the setting of the BLNS23 keyed-verification instantiation [1]. Let R_q be its ring, with dimension $d = 64$ and $\lceil \log_2 q \rceil = 152$. Vectors of $n = 93$ ring elements are written in bold, as are matrices. The rounding map $\lceil \cdot \rceil_p : R_q \rightarrow R_p$ with $p = 4$ keeps the high bits of each coefficient. A 32-byte seed z is expanded by an extendable-output function (XOF) into pseudorandom bytes; a fixed sampling procedure turns those bytes into messages and bounded-distribution vectors. Labels fed to hash functions are distinct strings that separate the different uses.

A *lane* is an index $i \in \{0, \dots, \kappa - 1\}$. A *branch* is one of two candidates $b \in \{0, 1\}$ prepared in a lane. A *session* is one issuance attempt, identified by a session identifier sid that names the protocol version, the issuer, and the key epoch, and carries fresh values from both parties.

Games, actors, and advantages. We use the game-playing convention. The *actors* are the user $\mathbf{U}$, the signer $\mathbf{S}$, and the verifier $\mathbf{V}$, given as (interactive) algorithms; $\mathcal{A}$ denotes an adversary, and $\mathcal{B}$, $\mathcal{D}$ denote reductions built from it. A *game* $\mathbf{G}$ is a probabilistic experiment run with an adversary; we write $\Pr[\mathbf{G}_{\mathcal{A}}(\lambda) \Rightarrow b]$ for the probability that the experiment halts with output $b \in \{0, 1\}$, taken over the random oracles, the challenger’s coins, and $\mathcal{A}$ ’s coins. The security parameter is λ , with $\kappa = \kappa(\lambda)$; all algorithms run in time polynomial in λ unless stated otherwise, and attack costs are stated as query counts. The *advantage* of $\mathcal{A}$ in distinguishing two games $\mathbf{G}^0, \mathbf{G}^1$ is

$$\text{Adv}_{\mathcal{A}}^{0/1}(\lambda) = |\Pr[\mathbf{G}_{\mathcal{A}}^0(\lambda) \Rightarrow 1] - \Pr[\mathbf{G}_{\mathcal{A}}^1(\lambda) \Rightarrow 1]|.$$

Query budgets are named as they appear: Q_h challenge-hash queries, Q accepted issuance sessions, Q_R total random-oracle queries, and q_V verification attempts.

3 The construction

Formally, the scheme is a tuple of algorithms $(\text{KeyGen}, \mathbf{U}_1, \mathbf{S}, \mathbf{U}_2, \mathbf{V})$. The signer runs **KeyGen**: it samples the public matrix B , the secret $(\mathbf{s}, \mathbf{e}_1)$ with $\|\mathbf{s}\| \leq \beta_s$ and $\|\mathbf{e}_1\| \leq \beta_e$, and a key K for the error derivation, and publishes $\text{pk} = (B, \mathbf{t})$ with $\mathbf{t}^T = \mathbf{s}^T B + \mathbf{e}_1^T$. Issuance is the two-message interaction

$$\text{msg}_1 \leftarrow \mathbf{U}_1(\text{pk}, \text{sid}), \quad (h, \pi_{\text{resp}}) \text{ or } \perp \leftarrow \mathbf{S}(\text{sk}, \text{msg}_1), \quad \sigma \text{ or } \perp \leftarrow \mathbf{U}_2(\text{st}_{\mathbf{U}}, h, \pi_{\text{resp}}),$$

and presentation is checked by $\mathbf{V}(\text{sk}, \sigma) \in \{0, 1\}$. The four subsections below define $\mathbf{U}_1$, the challenge derivation, $\mathbf{S}$, and the pair $(\mathbf{U}_2, \mathbf{V})$ in turn.

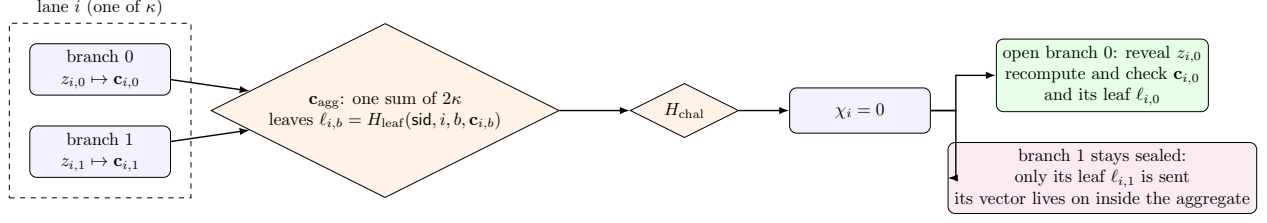


Figure 1: One lane under the aggregate commitment. Both branches enter the single sum $\mathbf{c}_{\text{agg}}$ and the leaf list; the challenge is taken over both, so the branch table is fixed before χ exists. The challenge bit χ_i elects which branch is opened and checked. The other branch is never transmitted or checked: only its leaf hash is sent, and its vector remains inside the aggregate. Roles reverse when $\chi_i = 1$.

3.1 Preparing the branches

The user algorithm $\mathbf{U}_1$ independently constructs two branches in each lane. It samples a 32-byte seed $z_{i,b}$ and expands it:

$$(m_{i,b}, \mathbf{r}_{i,b}, \mathbf{e}_{2,i,b}) = G(\text{sid}, i, b, z_{i,b}),$$

where $m_{i,b}$ is 32 bytes and the two vectors have the prescribed bounded distributions. It computes

$$\begin{aligned} \rho_{i,b} &= H_\rho(\text{sid}, i, b, \mathbf{r}_{i,b}, \mathbf{e}_{2,i,b}), \\ \mathbf{u}_{i,b} &= H_R(\text{credential-lane}, i, m_{i,b}, \rho_{i,b}), \\ \mathbf{c}_{i,b} &= B \mathbf{r}_{i,b} + \mathbf{e}_{2,i,b} + \mathbf{u}_{i,b}, \\ \ell_{i,b} &= H_{\text{leaf}}(\text{sid}, i, b, \mathbf{c}_{i,b}). \end{aligned}$$

Three details are essential later. First, the tag $\rho_{i,b}$ binds the branch to its session: valid branches can never be reused across sessions. Second, the lane index i sits *inside* the lane hash H_R , so a branch prepared for lane i is meaningless in any other lane, even though all lanes share the same matrix B . Dropping the lane label would let an attacker move branches between lanes freely; it is mandatory here. Third, the *leaf hash* $\ell_{i,b}$ commits the branch to its position (i, b) and to the session before the challenge exists; because it covers sid , even content that is never opened cannot be replayed across sessions. Section 3.2 explains why the challenge must be taken over these leaves and over the aggregate alike.

3.2 One aggregate commitment and the challenge

$\mathbf{U}_1$ sums all 2κ branch vectors into one aggregate

$$\mathbf{c}_{\text{agg}} = \sum_{i=0}^{\kappa-1} (\mathbf{c}_{i,0} + \mathbf{c}_{i,1}) \in R_q^n,$$

and only then derives the challenge from the aggregate and the complete leaf list

$$\chi = (\chi_0, \dots, \chi_{\kappa-1}) = H_{\text{chal}}(\text{protocol}, \text{issuer}, \text{pk}, \text{sid}, \mathbf{c}_{\text{agg}}, \ell_{0,0}, \ell_{0,1}, \dots, \ell_{\kappa-1,0}, \ell_{\kappa-1,1}) \in \{0, 1\}^\kappa.$$

Both inputs are load-bearing, for different reasons. The leaf hashes fix the branch *table* — which vector occupies each position (i, b) — before the challenge exists. Without them the challenge would

commit only to a sum, and a sum does not determine its summands: the user could choose the decomposition of the aggregate into branches *after* seeing the challenge, and the per-lane soundness argument of Section 5.2 would not apply. The aggregate must additionally enter the challenge itself, because it is the value the signer peels: the leaves can never constrain it, since the signer never learns the unopened preimages and consistency of the aggregate with the table cannot be checked. If the challenge did not cover the aggregate, the user could fix the table, learn χ , and only then choose the aggregate so that the peeled sum equals a chosen vector — the key-recovery attack of Section 5.1 at constant cost. The encoding of the aggregate and the leaf list must be canonical and the domain separation labels fixed, so that the hash input has only one interpretation.

The first message output by U_1 is

$$\text{msg}_1 = (\text{sid}, \mathbf{c}_{\text{agg}}, \{z_{i,\chi_i}\}_{i=0}^{\kappa-1}, \{\ell_{i,1-\chi_i}\}_{i=0}^{\kappa-1}).$$

Note what is *not* sent: the unopened branch vectors. The opened leaves are recomputed from the seeds; the unopened leaves arrive as hashes only. The aggregate remains the only commitment to the *values*, but the leaf list is a commitment to the *table*: every branch, including junk that is never opened, is position-bound and session-bound before the challenge exists.

3.3 Peeling and the aggregate response

The signer algorithm S expands every revealed seed, reconstructs each opened branch, and computes its leaf hash. Together with the received unopened leaf hashes this reassembles the full leaf list, and S recomputes χ from the received aggregate and that list. It checks that each reconstructed branch was formed for this session and lane (labels, session identifier, lattice equation, size bounds, encoding), and reconstructs the opened set $\{\mathbf{c}_{i,\chi_i}\}_i$. If any check fails it returns $\perp$ without releasing any secret-dependent value. Otherwise it *peels* the opened branches off the aggregate:

$$\mathbf{C}_{\text{sel}} = \mathbf{c}_{\text{agg}} - \sum_{i=0}^{\kappa-1} \mathbf{c}_{i,\chi_i} = \sum_{i=0}^{\kappa-1} \mathbf{c}_{i,1-\chi_i}.$$

We call $b_i^* = 1 - \chi_i$ the *selected branch* in lane i and write $(m_i^*, \rho_i^*, \mathbf{r}_i^*, \mathbf{e}_{2,i}^*)$ for its contents. The signer never sees the selected vectors individually; it holds only their sum.

S derives a small deterministic error from its secret key K and the complete received input,

$$e = H_{D_e}(K, \text{sid}, \mathbf{c}_{\text{agg}}),$$

and answers with the aggregate response

$$h = \mathbf{s}^T \mathbf{C}_{\text{sel}} + e \in R_q,$$

together with one zero-knowledge proof of the single-key response relation

$$\mathcal{R}_{\text{resp}} : \begin{cases} h = \mathbf{s}^T \mathbf{C}_{\text{sel}} + e, \\ \mathbf{t}^T = \mathbf{s}^T B + \mathbf{e}_1^T, \\ \|\mathbf{s}\| \leq \beta_s, \|\mathbf{e}_1\| \leq \beta_e, \|e\| \leq \beta_e. \end{cases}$$

Because there is one key, the proof covers one key relation. These are linear lattice equations and norm bounds, provable with a LaBRADOR-style proof [3], and they avoid the hash evaluations that made the BLNS23 user proof expensive. The signer releases only h and the proof: partial sums, per-lane values, failure details, and timing must not leak secret-dependent information.

3.4 Finishing and presenting the credential

After checking the response proof, the user algorithm U_2 removes the blinding terms one branch at a time and rounds:

$$v = \left\lfloor h - \mathbf{t}^T \sum_{i=0}^{\kappa-1} \mathbf{r}_i^* \right\rfloor_p, \quad \sigma = ((m_i^*, \rho_i^*)_{i=0}^{\kappa-1}, v).$$

The selected seeds stay private even at presentation: a seed would reveal the blinding vectors and let the issuer match the credential to its issuance.

At presentation, the verifier algorithm V (which holds the same key) recomputes the expected value and accepts exactly when

$$v = \left\lfloor \mathbf{s}^T \sum_{i=0}^{\kappa-1} H_R(\text{credential-lane}, i, m_i^*, \rho_i^*) \right\rfloor_p.$$

The lane hash used at presentation must not contain the issuance session identifier; the session identifier is bound into branch formation and the challenge only. An application derives one payload from the ordered message/tag list:

$$M = H_{\text{final}}(\text{issuer}, \text{type}, (m_i^*, \rho_i^*)_{i=0}^{\kappa-1}).$$

4 Why honest credentials verify

The unblinding step cancels most of the noise. For an honestly generated selected branch,

$$\mathbf{s}^T \mathbf{c}_i^* - \mathbf{t}^T \mathbf{r}_i^* = \mathbf{s}^T \mathbf{u}_i^* + (\mathbf{s}^T \mathbf{e}_{2,i}^* - \mathbf{e}_1^T \mathbf{r}_i^*),$$

so

$$h - \mathbf{t}^T \sum_i \mathbf{r}_i^* = \mathbf{s}^T \sum_i \mathbf{u}_i^* + \underbrace{\sum_i (\mathbf{s}^T \mathbf{e}_{2,i}^* - \mathbf{e}_1^T \mathbf{r}_i^*)}_{\text{correctness error}} + e.$$

The verifier recomputes the main term from the presented messages and tags. Acceptance requires that the correctness error never crosses a rounding boundary. The per-branch error terms have the same shape as in BLNS23's own verification equation, but here they are summed over κ lanes: the error is a sum of $2\kappa + 1$ small terms, so its size grows with the lane count (about $\sqrt{\kappa}$ in the typical randomized model, κ in the worst case), and all terms share one secret $\mathbf{s}$ and one public-key error $\mathbf{e}_1$. Formally, correctness is the statement

$$\Pr[V(\text{sk}, \sigma) = 1 : \text{sk} \leftarrow \text{KeyGen}, \sigma \leftarrow \langle U(\text{pk}), S(\text{sk}) \rangle] \geq 1 - \varepsilon_{\text{corr}},$$

over the oracles and the parties' coins, with $\varepsilon_{\text{corr}}$ bounded by the rounding-failure analysis above.

5 Security analysis

We state the attacks we know, the design rules they force, the part of the security claim that can be proven in the random-oracle model (Section 5.5), and the isolation of the part that cannot (Section 5.6).

5.1 The load-bearing order: commit before challenge

The aggregate and the branch table must be fixed before the challenge exists. To see why, suppose for a moment the opposite: the user learns χ (or its effect on which branches are opened) before fixing $\mathbf{c}_{\text{agg}}$. Then the user can pad the aggregate with a junk term

$$J = \mathbf{w} - \sum_i \mathbf{c}_i, 1-\chi_i$$

so that the peeled value equals a *chosen* vector $\mathbf{w}$. Two immediate consequences:

- **Key recovery.** With $\mathbf{w}$ the j -th unit vector, the response is $h = s_j + e$: one secret-key coordinate plus small noise. A handful of sessions per coordinate, with fresh session identifiers decorrelating the noise, recovers the entire secret key.
- **Free splicing.** With $\mathbf{w}$ chosen to cancel the honest content of a previous session and inject copied content, the copying attacks of Section 5.7 cost a constant number of hash trials instead of an exponential search.

The construction forbids this by deriving χ from the full aggregate and the complete leaf list with a hash function modeled as a random oracle, and by including fresh signer randomness in sid . The leaf list is not optional here: if the challenge covered the leaves but not the aggregate, the aggregate itself could be chosen after χ is known, and the same padding attack would return at constant cost (Section 3.2). In a two-message protocol the signer’s fresh value must come from an authenticated session setup or a previously established context; without it, the user can precompute entire exchanges long before contact. The order carries decisive weight here, because it is the *only* barrier: with a single key, nothing algebraic separates one lane’s vectors from another’s (Section 5.10).

5.2 Hiding malformed branches

Call a branch *openable* if a seed regenerates it and it passes every formation check. The signer checks only opened branches, so an attacker that wants a malformed branch to affect the response must keep it unopened. Because the challenge covers the complete leaf list, the branch table is fixed before the challenge exists: for a committed table containing one openable and one malformed branch in each of k attacked lanes, a uniform challenge opens all k openable branches with probability 2^{-k} ; each Fiat–Shamir trial re-randomizes the challenge, so q trials succeed with probability at most about $q2^{-k}$. Lemma 5.1 in Section 5.5 is the exact form of this bound.

The leaf binding is what makes this argument valid, and it repairs a real defect of the bare aggregate. A sum does not determine its summands: if the challenge were taken over $\mathbf{c}_{\text{agg}}$ alone, no branch table would be committed at all, and the attacker could choose the decomposition of the aggregate into branches *after* computing the challenge — presenting whichever precomputed openable branches suit the realized χ and treating everything else as junk. The probability experiment above would then be the wrong one: per hash query the attacker would succeed whenever *any* favorable decomposition exists, a strictly larger event whose analysis resembles a list-decoding problem rather than κ independent coin flips. The leaf hashes eliminate this freedom at a cost of 16 KiB, and because each leaf covers sid , even never-opened junk is session-bound and cannot be replayed across sessions.

The essential point stands: an attacker may place *arbitrary* junk in unopened positions — the peeled sum is never checked — but any junk must now be committed, per position and per session,

before the challenge is known, and the challenge decides which honest branches remain in the sum. The next subsection explains why this makes the junk hard to aim.

5.3 Why chosen inputs seem out of reach: the fixpoint wall

The dangerous attacks above all need the peeled sum $\mathbf{C}_{\text{sel}}$ to land in a chosen shape: a unit vector for key recovery, or a previous session’s content for cancellation. The attacker controls the aggregate, so why can it not simply choose $\mathbf{C}_{\text{sel}}$?

Because the honest branches it is forced to include interfere in a challenge-dependent way. Every lane must contain at least one openable branch (otherwise the session is rejected with probability one), and the openable branches are honestly formed, session-bound vectors whose values are fixed once the leaf list is fixed. The peeled sum is

$$\mathbf{C}_{\text{sel}} = \underbrace{\sum_{i \in H} \mathbf{c}_{i, 1-\chi_i}}_{\text{honest, chosen by } \chi} + \underbrace{\sum_{i \in H^c} \mathbf{w}_i}_{\text{junk, committed before } \chi},$$

where H is the set of lanes whose selected branch is honest. To aim the sum, the attacker must choose the junk to cancel the honest part — but the honest part is a pseudorandom function of the challenge, and the challenge is a pseudorandom function of the aggregate, the junk included. The dependency is circular: choosing J fixes $\mathbf{c}_{\text{agg}}$, which fixes χ , which fixes the honest selected sum, which determines the J that would have been needed (Figure 3).

Resolving the cycle is a guess-and-check search. Guessing the full challenge pattern costs about 2^κ trials (or $2^{\kappa/2}$ with generic quantum search). A meet-in-the-middle variant that only asks for a chosen *difference* between two sessions’ peeled sums — enough for the key-recovery attack — costs about $2^{\kappa/2}$ classical trials. We know no cheaper route. This motivates the property on which everything here rests, stated informally here and proven for the challenge game in Section 5.5:

Structured-sum fixpoint hardness (informal statement; formal version and proof in Section 5.5). Consider an attacker that may prepare branch tables freely, make q random-oracle queries, and run Q issuance sessions against the construction of Section 3. Then, except with probability on the order of $q 2^{-\kappa/2}$, every accepted session’s peeled sum $\mathbf{C}_{\text{sel}}$ is *uncontrolled*: its honest content is a set of honestly formed, session-fresh, hash-derived branch vectors whose selection is decided by the challenge only after the table is committed, and no two accepted sessions’ peeled sums satisfy a linear relation committed before the later session’s challenge query.

Given this property, the responses $h = \mathbf{s}^T \mathbf{C}_{\text{sel}} + e$ are learning-with-errors samples whose public vectors the attacker can influence but not aim. Whether such samples are safe — and what exactly must be assumed to conclude that they are — is the subject of Section 5.6, which isolates the step as one decisional assumption (structured-sum HMLWE) and proves the links around it.

5.4 Why not a sum-to-target attack?

The most direct attempt on the fixpoint wall treats the aggregate as a sum to be solved: grind seeds to build a list of candidate vectors at every position, then search for a table whose unopened branches sum to a chosen target,

$$\sum_{i=0}^{\kappa-1} \mathbf{c}_{i, 1-\chi_i} = \mathbf{w}, \quad \chi = H_{\text{chal}}(\text{sid}, \mathbf{c}_{\text{agg}}, \{\ell_{i,b}\}),$$

or equivalently choose $\mathbf{c}_{\text{agg}}$ adversarially and search for opened branches summing to $\mathbf{c}_{\text{agg}} - \mathbf{w}$. Two constraints stack, and each repays careful reading.

The vector equality is exact, and the space is enormous. Exactness is forced: key recovery needs $\mathbf{C}_{\text{sel}}$ to isolate key coordinates, splicing needs previous sessions’ lane values reproduced exactly because the presentation check is over exact hash images, and an approximate hit $\mathbf{w} + \delta$ does not isolate anything, since $\mathbf{s}^T \delta$ is secret-dependent. The target is therefore one value in a space of bit-length $N = nd \lceil \log_2 q \rceil = 93 \cdot 64 \cdot 152 \approx 904,704$ bits. Sum-to-target over per-lane lists is the generalized birthday problem, solved by Wagner’s k -tree algorithm [4] — and its cost scales with the bit-length of the sum space, not with the security parameter: with κ lists it uses time and list size about $2^{N/(1+\lceil \log_2 \kappa \rceil)}$. At $\kappa = 512$ this is about $2^{90,470}$: the per-lane lists alone would require grinding that many seeds per lane. Even the information-theoretic minimum for a solution to exist is $\kappa t \geq N$ for lists of size 2^t , i.e. $t \approx 1,767$ bits of seed grinding per position. Exact aiming is not a 128-bit problem but a 90,000-bit one; the size of the ring — the very thing that makes the commitment vectors 113 KiB each — is what protects the construction. And the challenge constraint stacks on top: a table that solved the sum would still need χ to select the intended complement.

Why the working attacks look different. The balanced splice (Section 5.7) and the meet-in-the-middle difference attack described above never aim at a prescribed target: they manufacture their targets from copied content, and committing a junk pair with a chosen per-lane difference costs nothing. The N -bit constraint never bites; only the challenge-hiding cost remains, $2^{\kappa/2}$ classical or $2^{\kappa/4}$ quantum. The fixpoint property is, at heart, the claim that no middle path exists — no relaxation of exact aiming that stays useful to the attacker yet becomes feasible.

Caveats. Two qualifications keep this honest. First, relaxations: if a *set* T of dangerous peeled sums is exploitable — say, all sufficiently sparse differences rather than only unit vectors — the effective attack cost divides by $|T|$, and list-merging is exactly how an attacker would harvest relaxed targets. Bounding $|T|$ rigorously is priced by Lemma 5.7: every bit of $\log_2 |T|$ comes directly off the margin. Second, the argument is parametric: it rests on $N \gg \kappa(1 + \log_2 \kappa)$. Shrinking the ring, the vector length, or the challenge length to save bandwidth moves the construction toward the regime where $2^{N/(1+\log_2 \kappa)}$ crosses below $2^{\kappa/2}$ and sum-to-target attacks become the cheapest ones.

5.5 A random-oracle proof of the fixpoint game

We now prove the precise content of the boxed statement for the challenge game, in the classical random-oracle model. The proof covers everything the challenge hash is responsible for — hiding, aiming, and committed relations between sessions. What it does not cover is stated at the end.

The game. The *fixpoint experiment* $\text{Exp}_{\mathcal{A}}^{\text{fix}}(\lambda)$ runs between a challenger (playing the signer) and an adversary $\mathcal{A}$:

1. **Setup.** The challenger samples independent random oracles $G, H_\rho, H_R, H_{\text{leaf}}, H_{\text{chal}}, H_{De}$, runs KeyGen, hands $\mathcal{A}$ the public key, and enforces session-identifier uniqueness.
2. **Interaction.** $\mathcal{A}$ may query the random oracles and, at any time, submit an issuance message msg_1 ; the challenger runs $S(\text{sk}, \text{msg}_1)$ and returns its output (which may be $\perp$).

3. **Budgets.** Write Q_h for the number of challenge-oracle queries (a session submitted without such a query is answered at signing time and counted as one), Q for the number of accepted sessions, and Q_R for the total number of random-oracle queries.

Call a vector $\mathbf{c} \in R_q^n$ *openable at* (sid, i, b) relative to the attacker's query transcript if the transcript contains the full chain: a seed z with $G(\text{sid}, i, b, z) = (m, \mathbf{r}, \mathbf{e}_2)$ inside the norm bounds, the tag $\rho = H_\rho(\text{sid}, i, b, \mathbf{r}, \mathbf{e}_2)$, the image $\mathbf{u} = H_R(\text{credential-lane}, i, m, \rho)$, and the equation $\mathbf{c} = B\mathbf{r} + \mathbf{e}_2 + \mathbf{u}$. A *committed input* is a tuple $X = (\text{sid}, \mathbf{c}_{\text{agg}}, \ell_{0,0}, \ell_{0,1}, \dots, \ell_{\kappa-1,0}, \ell_{\kappa-1,1})$; write $\chi(X) = H_{\text{chal}}(\dots, X)$. For each position (i, b) the attacker either knows a preimage $\mathbf{c}_{i,b}$ with $H_{\text{leaf}}(\text{sid}, i, b, \mathbf{c}_{i,b}) = \ell_{i,b}$ or it does not; a position is *open* if a preimage is known and openable at (sid, i, b) . A lane is *free* if both positions are open, *risky* if exactly one is, and *dead* if neither is. Write $\text{acc}(X, \chi)$ for the event that the challenge position χ_i is open in every lane and the attacker produces the corresponding seeds; the peeled sum is then

$$\mathbf{C}_{\text{sel}}(X, \chi) = \mathbf{c}_{\text{agg}} - \sum_{i=0}^{\kappa-1} \mathbf{c}_{i, \chi_i},$$

well defined for acceptable χ because the opened preimages are openable, hence unique up to random-oracle collision.

Two features of the model do all the work. First, $\chi(X)$ is uniform in $\{0, 1\}^\kappa$ and independent of everything the attacker has seen before the query at X . Second, the number of committed inputs and openable chains the attacker can produce is bounded by its query count.

We condition throughout on two global events whose failure probabilities are negligible at the parameters of Section 6: (i) *oracle regularity*: no collisions occur in any random oracle, and no openable chain exists that the attacker did not query — failure probability $O(Q_R^2 2^{-256})$; (ii) *distinctness*: within each committed input, the values $\mathbf{C}_{\text{sel}}(X, \chi)$ over acceptable χ are distinct. A collision is an equation $\sum_{i \in S} \pm(\mathbf{u}_{i,0} - \mathbf{u}_{i,1}) = (\text{lattice parts})$ in the H_R outputs for some nonempty $S \subseteq \{0, \dots, \kappa - 1\}$; a union bound over committed inputs, subsets S , and tuples of H_R queries gives failure probability at most $2^\kappa Q_R^{2\kappa+1} 2^{-N}$ with $N = n d \lceil \log_2 q \rceil \approx 904,704$, which is negligible for any $Q_R \ll 2^{N/(2\kappa)} \approx 2^{883}$.

The experiment's three failure events are

Bad_{hide} : some accepted session has at most $\kappa/2$ free lanes,

Bad_{tgt} : some accepted session's peeled sum lands in a target set committed before its challenge query,

Bad_{rel} : two accepted sessions' peeled sums satisfy a relation committed before the later session's challenge query.

Lemma 5.1 (commit-then-challenge soundness). *For a committed input with r risky lanes and no dead lane, $\Pr_\chi[\text{acc}(X, \chi)] = 2^{-r}$; a dead lane makes acceptance impossible. Consequently*

$$\Pr[\text{Bad}_{\text{hide}}] \leq Q_h 2^{-\kappa/2}.$$

Proof. Acceptance requires χ_i to name an open position in every lane, and under the uniform χ the lanes are independent: free lanes always accept, risky lanes accept with probability $1/2$, dead lanes never. The first claim follows. An accepted session with fewer than $\kappa/2$ free lanes has at least $\kappa/2$ risky lanes, so per committed input the probability is at most $2^{-\kappa/2}$; the bound is a union over the at most Q_h committed inputs. $\square$

Lemma 5.1 is the quantitative form of Section 5.2. Its content for what follows: outside Bad_{hide} , every accepted session has more than $\kappa/2$ free lanes, and its peeled sum can be written in the affine

form

$$\mathbf{C}_{\text{sel}}(X, \chi) = \mathbf{C}_0(X) + \sum_{i \in F} \chi_i \Delta_i, \quad \Delta_i = \mathbf{c}_{i,0} - \mathbf{c}_{i,1},$$

where F is the free-lane set, $|F| > \kappa/2$, and every $\mathbf{c}_{i,b}$ is openable — honestly formed and bound to (sid, i, b) . At commit time χ is uniform and unknown, so the peeled sum is an affine image of the challenge over more than $\kappa/2$ committed honest differences. This is the precise form of “uncontrolled”: the attacker learns χ immediately after its query, so unpredictability holds only at commit time, and changing the selection requires a new query.

Lemma 5.2 (target avoidance). *Fix a committed input X and a target set $T \subseteq R_q^n$, both chosen before the challenge query at X (they may depend arbitrarily on the attacker’s transcript so far). Then, under distinctness,*

$$\Pr_{\chi}[\text{acc}(X, \chi) \wedge \mathbf{C}_{\text{sel}}(X, \chi) \in T] \leq |T| 2^{-\kappa}, \quad \text{hence} \quad \Pr[\text{Bad}_{\text{tgt}}] \leq Q_h |T| 2^{-\kappa}.$$

Proof. Acceptance is possible for at most $2^{|F|}$ challenge patterns, and by distinctness the map $\chi \mapsto \mathbf{C}_{\text{sel}}(X, \chi)$ is injective on them, so at most $|T|$ acceptable patterns land in T . The challenge is uniform over 2^κ patterns. $\square$

For key recovery, T is the set of peeled sums from which key material is extractable — unit vectors, and whatever sparse or structured vectors suffice — and the lemma prices the attack at $Q_h |T| 2^{-\kappa}$. Every bit of $\log_2 |T|$ comes directly off the margin; Lemma 5.7 below extends the same pricing to target sets the attacker does not commit but selects by rejection.

Lemma 5.3 (committed-relation resistance). *Let session 1 be accepted with peeled sum $\mathbf{C}_1$. For any later committed input X_2 and any relation set $T_{\text{rel}} \subseteq R_q^n$ committed together with X_2 — after session 1 completes, but before the challenge query at X_2 —*

$$\Pr[\text{acc}(X_2, \chi(X_2)) \wedge \mathbf{C}_2 - \mathbf{C}_1 \in T_{\text{rel}}] \leq |T_{\text{rel}}| 2^{-\kappa}$$

under distinctness. Union over inputs and prior sessions: $\Pr[\text{Bad}_{\text{rel}}] \leq Q_h Q |T_{\text{rel}}| 2^{-\kappa}$.

Proof. Identical to Lemma 5.2 with $T = \mathbf{C}_1 - T_{\text{rel}}$: at most $|T_{\text{rel}}|$ acceptable patterns for $\chi(X_2)$ satisfy the relation, and $\chi(X_2)$ is uniform and independent of the attacker’s view at commit time, session 1 included. $\square$

Theorem 5.4 (structured-sum fixpoint hardness, classical ROM). *Let the signer enforce session-identifier uniqueness, and let $|T|$ and $|T_{\text{rel}}|$ be the largest target and relation sets the attacker commits. Then*

$$\Pr[\text{Bad}_{\text{hide}} \vee \text{Bad}_{\text{tgt}} \vee \text{Bad}_{\text{rel}}] \leq Q_h 2^{-\kappa/2} + Q_h |T| 2^{-\kappa} + Q_h Q |T_{\text{rel}}| 2^{-\kappa} + 2^\kappa Q_R^{2\kappa+1} 2^{-N} + O(Q_R^2 2^{-256}),$$

and outside this event every session the attacker completes satisfies: (i) its table has more than $\kappa/2$ free lanes, so its peeled sum is an affine image of the challenge over more than $\kappa/2$ honestly formed, session-fresh, hash-derived branch differences committed before the challenge existed; (ii) its peeled sum avoids every target set committed before the challenge query; (iii) its peeled sum stands in no committed relation to the peeled sum of any previously accepted session.

Proof. Union bound over Lemmas 5.1–5.3 and the two conditioning events. $\square$

Tightness. The bounds are matched by the attacks of the preceding subsections. Hiding k malformed branches costs 2^k trials (Lemma 5.1). Aiming at a fixed target with $|T| \approx 1$ costs 2^κ (Lemma 5.2). A committed difference between two sessions costs $Q_h Q 2^{-\kappa} \approx 1$ at $Q_h \approx Q \approx 2^{\kappa/2}$ (Lemma 5.3) — exactly the meet-in-the-middle and balanced splice figures. The $2^{\kappa/2}$ floor is therefore not an artifact of missing analysis: it is where the proven bounds stop protecting the construction.

What is not covered. Three things. First, quantum query access: the proofs sample χ classically at query time and take union bounds over classical query transcripts; a quantum attacker queries in superposition, and the analysis needs compressed-oracle or equivalent techniques. The $2^{\kappa/4}$ quantum floor of Section 5.7 is only a generic-search estimate. Second, the step from uncontrolled peeled sums to lattice hardness: Theorem 5.4 says the attacker cannot aim $\mathbf{C}_{\text{sel}}$, not that $h = \mathbf{s}^T \mathbf{C}_{\text{sel}} + e$ with uncontrolled $\mathbf{C}_{\text{sel}}$ is safe. That step is taken up in Section 5.6, which isolates it as a single decisional assumption (structured-sum HMLWE), proves the links from it to key recovery and one-more unforgeability, and shows why no reduction to plain MLWE applies; the assumption itself remains open. Third, the simulation details of the one-more accounting: the counting argument is Theorem 5.8, but its hybrid steps — response-proof simulation, key validation, correctness — remain uninstantiated.

5.6 The lattice step: isolating the remaining assumption

Theorem 5.4 leaves a precise gap, and this subsection determines exactly what fills it. Condition on its good event. Every accepted session hands the attacker a sample

$$(\mathbf{C}_{\text{sel}}, h = \mathbf{s}^T \mathbf{C}_{\text{sel}} + e), \quad \mathbf{C}_{\text{sel}} = \mathbf{C}_0(X) + \sum_{i \in F} \chi_i \Delta_i, \quad |F| > \kappa/2,$$

in which *every* vector on the right-hand side is known to the attacker: it built the branch table, junk included. The only quantity fixed after commitment is the challenge, uniform over at least $|F|$ effective bits of selection. Issuance is therefore an oracle for learning-with-errors samples whose matrices the attacker knows completely and influences in exactly two ways: it chooses the table, and it chooses which challenge queries to carry to completion. The *lattice step* is the claim that such samples neither reveal $\mathbf{s}$ nor can be produced without it. We isolate that claim as one decisional assumption, prove the links from it to key-recovery resistance and one-more unforgeability, price the residual control the attacker retains over the matrices, and explain why no reduction to a standard assumption is available along known lines.

Why no standard reduction applies. A reduction must answer issuance queries without knowing $\mathbf{s}$. Both classical routes die on the construction’s two defining features. First, the single key: with independent lane keys, the reduction embeds its challenge in one lane’s key and computes the other lanes’ contributions locally, absorbing arbitrary content there; here every term of $\mathbf{C}_{\text{sel}}$ lives under the unknown $\mathbf{s}$, and no known-key lane exists to absorb anything. Second, the peeled sum is not a hash image: an HMLWE-style oracle (as in BLNS23’s Lemma 4.3) evaluates $\mathbf{s}$ only at fresh outputs of H_R , but $\mathbf{C}_{\text{sel}}$ contains committed junk $\mathbf{w}$ that never passed through H_R . Extraction is not the obstacle — the reduction can recover $\mathbf{w}$ from the attacker’s H_{leaf} query history — but no oracle will supply $\mathbf{s}^T \mathbf{w}$. Knowing the key forfeits the embedded challenge; not knowing it forfeits the junk term. Repairing either route means arranging for every term of the peeled sum to be a certified hash image, that is, proving well-formedness of the unopened branches — the very proof cut-and-choose exists to eliminate. The remaining assumption must therefore be made explicit.

Structured-sum HMLWE (ssHMLWE). The two games $\mathbf{G}_{\mathcal{A}}^{\text{real}}(\lambda)$ and $\mathbf{G}_{\mathcal{A}}^{\text{ideal}}(\lambda)$ (Game 4) differ only in the answer to an accepted session j : the real game returns $h_j = \mathbf{s}^T \mathbf{C}_{\text{sel},j} + e_j$; the ideal game returns an independent uniform $h_j \leftarrow \$ R_q$. The assumption: for every classical probabilistic polynomial-time adversary $\mathcal{A}$,

$$\text{Adv}_{\mathcal{A}}^{\text{ssHMLWE}}(\lambda) = |\Pr[\mathbf{G}_{\mathcal{A}}^{\text{real}}(\lambda) \Rightarrow 1] - \Pr[\mathbf{G}_{\mathcal{A}}^{\text{ideal}}(\lambda) \Rightarrow 1]| \leq \varepsilon_{\text{fix}} + \text{negl}(\lambda),$$

where ε_{fix} collects the error terms of Theorem 5.4.

$\mathbf{G}_{\mathcal{A}}^{\text{real}}(\lambda)$	$\mathbf{G}_{\mathcal{A}}^{\text{ideal}}(\lambda)$
sample oracles $G, H_\rho, H_R, H_{\text{leaf}}, H_{\text{chal}}, H_{D_e}$	same
$(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}$; enforce sid uniqueness	same
$b' \leftarrow \mathcal{A}^{H, S(\text{sk}, \cdot)}(\text{pk})$, where on accepted session j :	$b' \leftarrow \mathcal{A}^{H, S'(\text{sk}, \cdot)}(\text{pk})$, where:
$h_j := \mathbf{s}^T \mathbf{C}_{\text{sel},j} + e_j, \quad e_j = H_{D_e}(K, \text{sid}_j, \mathbf{c}_{\text{agg},j})$	$h_j \leftarrow \$ R_q$, independent of j
return $b' \in \{0, 1\}$	return $b' \in \{0, 1\}$

Figure 4: The ssHMLWE games. S' answers like S except that the aggregate response of every accepted session is replaced by an independent uniform ring element; opening checks, rejections, and the response-proof interface are unchanged.

Three remarks delimit the assumption. First, it is purely decisional: it asserts pseudorandomness of the response stream and says nothing about forgery by itself. Second, the error terms are load-bearing, and they are exactly the right ones: grinding two sessions to *equal* peeled sums would distinguish the games (then $h_1 - h_2 = e_1 - e_2$ is small), and Lemma 5.2 with $|T| = 1$ prices that at $2^{-\kappa}$ per query pair; the known attacks at the $2^{\kappa/2}$ frontier are forgeries and meet-in-the-middle searches, not distinguishers. No attack on ssHMLWE itself is known at any budget. Third, the assumption is falsifiable in a precise sense made explicit after Lemma 5.7 below.

Key recovery reduces to the assumption. The first link needs no lattice machinery at all, because a candidate key can be checked against the attacker’s own transcript. Define the key-recovery experiment $\mathbf{Exp}_{\mathcal{A}}^{\text{kr}}(\lambda)$: run $\mathcal{A}$ in the real issuance game; $\mathcal{A}$ wins if it outputs $\hat{\mathbf{s}} = \mathbf{s}$.

Theorem 5.5 (keys are self-verifying). *For every adversary $\mathcal{A}$ with $\Pr[\mathbf{Exp}_{\mathcal{A}}^{\text{kr}}(\lambda) \Rightarrow 1] = \delta$, there is an ssHMLWE distinguisher $\mathcal{D}$ with $\text{Time}(\mathcal{D}) \leq \text{Time}(\mathcal{A}) + O(Q)$ inner products and*

$$\text{Adv}_{\mathcal{D}}^{\text{ssHMLWE}}(\lambda) \geq \delta - \left(\frac{2\beta_e + 1}{q} \right)^d,$$

where β_e is the response-error bound of Section 3.3.

Proof. $\mathcal{D}$ relays $\mathcal{A}$ ’s issuance sessions to its own challenger and reads every $\mathbf{C}_{\text{sel}}$ off the transcript. When $\mathcal{A}$ outputs a candidate $\hat{\mathbf{s}}$, the distinguisher outputs 1 (“real”) if and only if $\|h_j - \hat{\mathbf{s}}^T \mathbf{C}_{\text{sel},j}\|_\infty \leq \beta_e$ for the first accepted session j . In the real game a correct key passes with probability one, since $h_j - \mathbf{s}^T \mathbf{C}_{\text{sel},j} = e_j$ and $\|e_j\| \leq \beta_e$ is enforced by the response proof. In the ideal game h_j is uniform in R_q and independent of $\hat{\mathbf{s}}$, so each of its d coefficients falls in the required window with probability $(2\beta_e + 1)/q$. The bound follows; at the parameters of Section 6 the subtracted term is below 2^{-128} . (If $\mathcal{A}$ outputs an *alternative* key consistent with the public key, the difference $\mathbf{s} - \hat{\mathbf{s}}$ is a short kernel element of B , and passing the check on every session additionally requires $(\mathbf{s} - \hat{\mathbf{s}})^T \mathbf{C}_{\text{sel},j}$ to be small on the attacker’s own samples — itself an ssHMLWE-type event.) $\square$

The content of Theorem 5.5 is that key recovery is *not* the hard direction: any key extractor is a distinguisher, with no search-to-decision lemma and no loss. All remaining content of the lattice step lives in the other direction — producing or predicting responses — and in distinguishing strategies that do not recover the key. The next two lemmas constrain the second class.

Lemma 5.6 (closed cycles cost a search or a SIS solution). *Consider a distinguisher $\mathcal{A}$ that computes, from its final view, short coefficients $(\alpha_1, \dots, \alpha_Q) \in R_q^Q$, not all zero, and outputs 1 when $\|\sum_j \alpha_j h_j\|$ is small. In the ideal game this sum is uniform, so $\mathcal{A}$'s advantage is bounded by the probability that the real-game sum is small. Writing $\mathbf{v} = \sum_j \alpha_j \mathbf{C}_{\text{sel},j}$, exactly one of the following holds:*

- (i) $\mathbf{v} \neq 0$. The test then compares $\mathbf{s}^T \mathbf{v} + \sum_j \alpha_j e_j$ against uniform for a known $\mathbf{v}$: a decisional keyed-evaluation guess whose hardness is the ssHMLWE assumption itself, not an attack on it.
- (ii) $\mathbf{v} = 0$, and the relation was fixed before the last involved challenge query. Then $\Pr[\mathcal{A} \text{ succeeds}] \leq Q_h Q |T_{\text{rel}}| 2^{-\kappa}$ by Lemma 5.3, with $|T_{\text{rel}}|$ counting the relations the attacker committed to.
- (iii) $\mathbf{v} = 0$, and the relation was found only after all involved challenges were sampled. Then $\mathcal{A}$ has produced a short kernel vector of the collected matrix $A = [\mathbf{C}_{\text{sel},1} \mid \dots \mid \mathbf{C}_{\text{sel},Q}]$: a module-SIS solution on samples it could not aim.

Proof sketch. The trichotomy is exhaustive by the definition of $\mathbf{v}$ and by whether the coefficients were determined before or after the last involved challenge query. Case (i) restates the test. Case (ii) is Lemma 5.3 applied to the last involved session, conditioned on all earlier ones. Case (iii) names the residual computation: given the full matrix A , find short $\boldsymbol{\alpha} \neq 0$ with $A\boldsymbol{\alpha} = 0$. $\square$

The point of Lemma 5.6 is that the natural linear distinguisher — cancel the keyed terms exactly and test the residual noise — has no free instance here. Engineering the cancellation into the samples is a committed relation and pays the proven frontier; discovering it afterwards is generic module-SIS. What remains is the attacker's one real degree of freedom: choosing which samples to keep.

Lemma 5.7 (residual control is priced by density). *Condition on the good events of Theorem 5.4. Suppose $\mathcal{A}$ keeps a session only when its peeled sum lands in a set T chosen from its view — possibly after the challenge query, as rejection sampling. Then the expected number of challenge queries per kept session is at least $2^\kappa/|T|$:*

$$\mathbb{E}[\text{queries per kept session}] \geq \frac{2^\kappa}{|T|}.$$

Proof. Fix the committed input X . Under distinctness, the reachable set $R_X = \{\mathbf{C}_{\text{sel}}(X, \boldsymbol{\chi})\}$ over acceptable $\boldsymbol{\chi}$ has size $2^{|F|}$, and $\boldsymbol{\chi}$ is uniform over $\{0, 1\}^\kappa$, so one query lands in T with probability $p = |T \cap R_X| 2^{-\kappa} \leq |T| 2^{-\kappa}$. Discarding misses changes nothing per query, and each kept session therefore costs at least $1/p \geq 2^\kappa/|T|$ queries in expectation. $\square$

Lemma 5.7 prices every attempt to *condition* the collected matrices, and the prices land on both sides of the design floor in an instructive way. Aiming at unit vectors ($|T| \approx 2^{13}$ at the parameters of Section 6) costs about $2^{\kappa-13}$ queries per sample: the key-recovery route of Section 5.1 stays far above the $2^{\kappa/2}$ floor. Pinning one prescribed coefficient of $\mathbf{C}_{\text{sel}}$ to a chosen value costs $q \approx 2^{152}$ queries per sample — below the floor at $\kappa = 512$, and useless on its own: shrinking the module-LWE

dimension from $nd = 5,952$ to k' requires pinning $5,952 - k'$ coordinates in *every* collected sample, at cost $Q \cdot 2^{152(5952-k')}$. Between “one coordinate at 2^{152} , useless” and “thousands of coordinates, astronomically beyond the floor,” no graduated attack is known. Conditioning two sessions into a short difference — the raw material of a closed cycle — is priced separately by Lemma 5.3 at $|T_{\text{rel}}| 2^{-\kappa}$ per pair of queries.

What would break the assumption. Lemmas 5.6 and 5.7 locate the residual content of ssHMLWE precisely. The assumption fails exactly if there is an efficiently checkable property of the peeled sum that simultaneously (a) makes the resulting samples easy — key-extractable or linearly testable — and (b) is dense enough to afford at the attacker’s budget, or is reachable by solving module-SIS on the collected matrix. No such property is known, and the two naive families above are priced at or above the $2^{\kappa/2}$ floor (with the single, useless exception noted). This is the sense in which the assumption is falsifiable: exhibit a checkable, dense, key-extractable property of the peeled sum.

One-more unforgeability from the same assumption. With the common-message binding of Section 5.8, the one-more accounting also reduces to ssHMLWE, up to simulation details that remain uninstantiated. Formally, the *one-more experiment* $\mathbf{Exp}_A^{\text{omuf}}(\lambda)$ runs the issuance game with $\mathcal{A}$ and additionally gives it a verification oracle $\text{Check}(M, \{\phi_i, \rho_i\}_i, v) \in \{0, 1\}$ implementing V ’s computation. Writing Q for the number of accepted issuances and q_V for the number of Check queries, $\mathcal{A}$ wins if it presents $Q + 1$ tuples with pairwise distinct messages M that Check accepts.

Theorem 5.8 (one-more unforgeability, conditional). *Assume ssHMLWE, binding and public rerandomizability of the common-message commitment of Section 5.8, and the standard hybrid steps of the BLNS23 proof architecture (response-proof simulation, key validation, correctness). Then for every adversary $\mathcal{A}$ there exist $\mathcal{B}$ and $\mathcal{B}_{\text{bind}}$, of essentially the same running time, such that*

$$\Pr[\mathbf{Exp}_A^{\text{omuf}}(\lambda) \Rightarrow 1] \leq \text{Adv}_{\mathcal{B}}^{\text{ssHMLWE}}(\lambda) + \varepsilon_{\text{fix}} + \text{Adv}_{\mathcal{B}_{\text{bind}}}^{\text{bind}}(\lambda) + q_V 2^{-\nu} + \varepsilon_{\text{hyb}} + \varepsilon_{\text{corr}},$$

where ε_{fix} collects the error terms of Theorem 5.4 and ν is the min-entropy of the rounded value ($\nu = d \log_2 p = 128$ at the parameters of Section 6).

Proof sketch. Idealize every issuance response using ssHMLWE, losing $\text{Adv}_{\mathcal{B}}^{\text{ssHMLWE}} + \varepsilon_{\text{fix}}$. Thereafter lazy-sample the keyed function $x \mapsto \mathbf{s}^T H_R(x)$ as a random function at all fresh points; ε_{hyb} and $\varepsilon_{\text{corr}}$ cover the uninstantiated simulation and correctness steps named in the statement. Consider a presented credential for a message M' distinct from every issued message M_j . Commitment binding gives $\mu'_i = \text{Com}(M'; \phi'_i) \neq \text{Com}(M_j; \phi_i^{(j)})$ for every issued session j and every lane i , so each of the κ lane inputs (i, μ'_i, ρ'_i) is a fresh point of the random function, regardless of the tags. The required value is then the rounding of a sum of κ independent fresh random values — uniform, with min-entropy ν — and each presentation attempt succeeds with probability at most $2^{-\nu}$. $\square$

Status. After this subsection, the open core of the construction is one sentence long: the pseudo-randomness of the response stream (ssHMLWE), its quantum analogue, and the named simulation details. Everything else — the challenge game, the linear-attack surface, the pricing of residual control, key recovery, and the one-more counting — is now a theorem or a reduction. This is weaker than a proof under a standard assumption, and the difference is real: ssHMLWE is a new, protocol-specific assumption, and the barrier paragraph above explains why the standard route cannot reach it. But the assumption is minimal, decisional, and falsifiable, and its surrounding structure is proven.

5.7 Copying and combining responses

The rounded aggregate, together with the tags, forms a message authentication code. Write

$$F_i(x) = \mathbf{s}^T H_R(\text{credential-lane}, i, x), \quad Y(x_0, \dots, x_{\kappa-1}) = \sum_i F_i(x_i),$$

ignoring the small correctness error. Y is a sum of per-lane values, so it satisfies a *rectangle relation*: if the same valid input $x_i = (m_i, \rho_i)$ could appear in two sessions, three issued credentials on three corners of a rectangle of inputs would combine into a fourth, unissued credential by $Y_{10} + Y_{01} - Y_{00} = Y_{11}$. The combined correctness error grows only by a constant factor.

This is why session freshness is mandatory, not optional: $\rho_{i,b}$ contains a globally unique *sid*, so reusing a valid input across sessions requires a hash preimage. The leaf hashes reinforce this at the branch level: since $\ell_{i,b}$ also covers *sid*, not even unopened junk can be carried over from one session’s aggregate into another’s. The enforcement must cover concurrency, rollback, and distributed replay-state failures.

Fresh sessions leave the *balanced splice*. An attacker copies old lane vectors into malformed branches (copies need no valid seeds, since unopened branches are never checked), grinds the challenge so the copies stay hidden, and thereby reproduces previous sessions’ lane values inside new responses. Splitting the lanes into two halves, two grinding sessions at $2^{\kappa/2}$ each produce the second and third corners, and the fourth is free: about $2^{\kappa/2}$ classical work, or $2^{\kappa/4}$ with generic quantum search. Formally, the splice wins the one-more experiment $\mathbf{Exp}^{\text{omuf}}$ of Section 5.6 as soon as $Q_h Q 2^{-\kappa} \approx 1$, which is exactly the boundary at which Lemma 5.3 stops protecting the construction. This is the cheapest attack we know, and it sets the parameter floor: $\kappa = 512$ for a 128-bit post-quantum margin. Section 5.8 shows how to demote the splice’s yield from a fresh-message forgery to a duplicate of an already-issued credential; the floor stands even so, because key recovery through the fixpoint wall independently costs about $2^{\kappa/2}$.

Two honest remarks. First, the single key makes cancellation *algebraically* easy: any lane vector can cancel any other, because all values live under one $\mathbf{s}$. But the fixpoint wall of Section 5.3 forces the cancellation term to be committed before the challenge, and the challenge-dependent honest contributions re-impose one hidden-malformed-branch bit per canceled lane — so the attack cost ends up at $2^{\kappa/2}$. Second, the random-oracle analysis of Section 5.5 shows that this figure is not an artifact of missing analysis: $\Pr[\text{Bad}_{\text{rel}}]$ reaches 1 exactly at $Q_h Q 2^{-\kappa} \approx 1$, that is, at $2^{\kappa/2}$ — and the splice sits exactly there. What the analysis does not give by itself is a bound on the *uses* of such a relation below it; Section 5.6 supplies the pricing: committed relations are bounded by Lemma 5.3, and relations found after the fact are module-SIS instances (Lemma 5.6).

5.8 Binding every lane to one message

The splicing attacks combine lane values across sessions. Their output is a credential whose lanes come from different sessions — hence, potentially, a credential for a message vector that was never issued as a whole. The strengthening in this subsection compresses what such mixing can achieve: bind every lane of a session to one hidden message M . Despite arriving late in the exposition, it should be read as part of the recommended construction, not as an option. Without it, the cheapest attack we know (Section 5.7) produces a credential for a fresh, never-issued message vector at $2^{\kappa/2}$ work; with it, the same work produces only a duplicate of a credential that was already issued. The mechanism is also a prerequisite for the one-more accounting (Theorem 5.8).

Use a commitment scheme that hides M , cannot be opened to two messages, and can be publicly rerandomized. The user commits $\mu_0 = \text{Com}(M; \phi_0)$; each branch carries a rerandomization

$\mu_{i,b} = \text{Com}(M; \phi_0 + \Delta_{i,b})$ with $\Delta_{i,b}$ seed-derived; μ_0 is included in the challenge hash input; opened branches are checked to be correct rerandomizations; the lane hash becomes $H_R(\text{credential-lane}, i, \mu_{i,b}, \rho_{i,b})$. At presentation the user reveals M and the selected randomness, and the verifier rebuilds every μ_i^* . Formally, binding is the requirement that every adversary $\mathcal{A}$ has $\text{Adv}_{\mathcal{A}}^{\text{bind}}(\lambda) = \text{negl}(\lambda)$ against the experiment in which $\mathcal{A}$ wins by outputting $(\mu, M, \phi, M', \phi')$ with $M \neq M'$ and $\text{Com}(M; \phi) = \mu = \text{Com}(M'; \phi')$.

The effect: a presented credential is consistent with exactly one message. A spliced credential must then reuse an already-issued message, since its lanes can only be consistent if the spliced sessions shared M . The attack is demoted from “forge a new message” to “duplicate an existing message’s credential.” For applications whose authorization semantics live entirely in M , duplicates are worthless and the splicing line of attack is effectively closed; where credentials are counted, serialized, or rate-limited, duplicates remain valuable and the demotion does not suffice on its own.

What the verifier must store. The demotion has a concrete operational pay-off. Because a presented credential is consistent with exactly one message, duplicate and replay suppression collapses to one short record per presentation: M , or a hash of it. Any credential the splicing attacks can produce shares its M with an honestly issued one, so the number of distinct presentable messages stays bounded by the number of issuances, and a single-use check on M enforces exactly that bound. Without the binding the same protection is far more expensive and more fragile: a spliced credential is a *new* combination that matches no single stored credential, so the verifier would have to record every presented lane input (m_i^*, ρ_i^*) — κ values per credential — and reject any presentation that reuses a lane. The binding replaces that per-lane database with one hash per message.

M as nullifier. In the vocabulary of anonymous credentials, M plays the role of a *nullifier*: the unique public value whose reuse identifies a double presentation. Two caveats come with the terminology. First, a nullifier must be unique per *issuance*, not per payload: if two users hold credentials for the same application message (an age bound, a residency claim), a single-use check on M would make them block each other. The message should therefore carry a per-issuance serial or nonce, so that M identifies the credential instance rather than the shared attribute. Second, revealing M links all presentations of the same credential to each other and to the payload. That linkage is usually the point of a nullifier — it is what makes double presentation detectable — but it is a deliberate privacy decision, and cross-context unlinkability would need a scoped image such as $H(M, \text{scope})$, with the scope bound into the lane hash so that the verifier can still recompute the lane inputs.

Three honest qualifications. First, the demotion is a conditional theorem (Theorem 5.8): its counting argument anchors on the commitment binding — distinct messages force fresh lane commitments in every lane — and on the structured-sum HMLWE assumption of Section 5.6, whose simulation details remain uninstantiated. Second, the binding does nothing against key recovery through the fixpoint wall, which independently costs about $2^{\kappa/2}$: the parameter floor $\kappa = 512$ stands whether or not duplicates are harmful. Third, the semantic cost is real but small. The base construction already derives a single application payload $M = H_{\text{final}}(\text{issuer}, \text{type}, (m_i^*, \rho_i^*)_i)$ from the ordered lane list, so applications see one message either way; what is given up is only the ability to treat individual lanes as independently meaningful messages.

5.9 Blindness

The signer sees: the session identifier, one aggregate vector, the κ opened seeds with their reconstructed branches, and the κ leaf hashes of the unopened branches. It never sees the selected seeds, the selected blinding vectors, or the final rounded value; at presentation the user reveals only messages, tags, and v , and the presentation hash contains no session identifier. This subsection carries the dishonest-signer blindness proof through, in the style of Section 5.6: every step is proven in the classical random-oracle model except one, which is isolated as a single decisional assumption — *short-decomposition indistinguishability* (SDI) — and that assumption is also *necessary*: a decider breaks blindness on the spot (the tightness paragraph below). Unlike structured-sum HMLWE, SDI is close to a standard assumption, and the asymmetry has a clean source. The unforgeability adversary builds its own branch tables, and the committed junk inside its peeled sums is exactly what no standard oracle can evaluate under $\mathbf{s}$; the blindness adversary faces an *honest* user and knows the signing key, and what it cannot see is the user’s blinding — nothing else.

The one statistic that matters. Fix a completed session and a presented credential $\sigma = ((m_i^*, \rho_i^*)_{i=0}^{\kappa-1}, v)$, both known to the signer. The signer holds $\mathbf{C}_{\text{sel}} = \mathbf{c}_{\text{agg}} - \sum_i \mathbf{c}_{i, \chi_i}$ and recomputes $\mathbf{U}^* = \sum_i H_R(\text{credential-lane}, i, m_i^*, \rho_i^*)$. If σ was produced by this session, then

$$\mathbf{V} := \mathbf{C}_{\text{sel}} - \mathbf{U}^* = B \mathbf{R}^* + \mathbf{E}_2^*, \quad \mathbf{R}^* = \sum_{i=0}^{\kappa-1} \mathbf{r}_i^*, \quad \mathbf{E}_2^* = \sum_{i=0}^{\kappa-1} \mathbf{e}_{2,i}^*,$$

so $\mathbf{V}$ has a *short decomposition*: an image under the public matrix B of a width- κ sum of bounded vectors, plus a bounded error. If σ came from a different session, then $\mathbf{V} = B \mathbf{R}^* + \mathbf{E}_2^* + (\mathbf{U}_{\text{self}}^* - \mathbf{U}^*)$, and the difference of two independent uniform vectors masks $\mathbf{V}$ to uniform over R_q^n . Linking a credential to its issuance session is therefore exactly the decision problem *does this $\mathbf{V} \in R_q^n$ have a short decomposition?* The rest of the subsection shows that every other statistic available to the signer is either erased by rounding (the response error and the correctness noise, Lemma 5.10) or priced by random-oracle preimage resistance (the seeds, the tags, the leaves, Lemma 5.11), so that this decision problem is the whole of blindness.

The experiment. The *dishonest-signer blindness experiment* $\text{Exp}_{\mathcal{A}}^{\text{blind}}(\lambda)$:

1. $\mathcal{A}$, playing the signer, outputs a public key pk (subject to (KV) below) and two distinct session identifiers $\text{sid}_0, \text{sid}_1$.
2. The experiment runs the honest user $(\mathbf{U}_1, \mathbf{U}_2)$ with $\mathcal{A}$ in two issuance sessions on inputs $(\text{pk}, \text{sid}_0)$ and $(\text{pk}, \text{sid}_1)$ — concurrently, if $\mathcal{A}$ wishes — producing σ_0, σ_1 , each possibly $\perp$.
3. If either $\sigma_j = \perp$, $\mathcal{A}$ is handed $(\perp, \perp)$; otherwise the experiment samples $b \leftarrow \$ \{0, 1\}$ and hands $\mathcal{A}$ (σ_b, σ_{1-b}) .
4. $\mathcal{A}$ outputs b' and wins if $b' = b$; its advantage is $\text{Adv}_{\mathcal{A}}^{\text{blind}}(\lambda) = |\Pr[b' = b] - \frac{1}{2}|$.

Three conventions are part of the model; the requirements paragraph below shows that each is load-bearing.

- (KV) *Key validation.* The public matrix is derived from a public seed, $B = \text{Expand}_B(\text{seed}_B)$, with Expand_B modeled as a random oracle. The response proof is knowledge-sound; the experiment runs the extractor on $\mathcal{A}$ ’s first verifying response proof and checks $\mathbf{t}^T = \mathbf{s}^T B + \mathbf{e}_1^T$, the norm bounds, and that some component of the extracted $\mathbf{s}$ is a unit in R_q (so that $\mathbf{s}^T \mathbf{U}$ is uniform for uniform $\mathbf{U}$). Any failure aborts both sessions.

- (SID) *Session freshness, user side.* The honest user rejects a session identifier it has been offered before.
- (ABORT) *Both-or-neither delivery.* If either session fails — a rejected proof, a malformed response, a failed check — both credentials are replaced by $\perp$, and nothing about which session failed may leak, timing included.

The assumption.

Short-decomposition indistinguishability (SDI). Fix a width $w \in \{1, \dots, \kappa\}$. The games $\mathbf{G}_{\mathcal{A}}^{\text{struc}}(\lambda, w)$ and $\mathbf{G}_{\mathcal{A}}^{\text{unif}}(\lambda, w)$ (Game 5) differ only in the challenge vector: the structured game returns $\mathbf{y} = B\mathbf{R} + \mathbf{E}_2$ with $(\mathbf{R}, \mathbf{E}_2) = \sum_{i=1}^w (\mathbf{r}_i, \mathbf{e}_{2,i})$ a sum of w honestly distributed branch vectors; the uniform game returns $\mathbf{y} \leftarrow \$ R_q^n$. The assumption: for every width w and every classical probabilistic polynomial-time adversary $\mathcal{A}$,

$$\text{Adv}_{\mathcal{A}}^{\text{SDI}(w)}(\lambda) = |\Pr[\mathbf{G}_{\mathcal{A}}^{\text{struc}}(\lambda, w) \Rightarrow 1] - \Pr[\mathbf{G}_{\mathcal{A}}^{\text{unif}}(\lambda, w) \Rightarrow 1]| \leq \text{negl}(\lambda).$$

$\mathbf{G}_{\mathcal{A}}^{\text{struc}}(\lambda, w)$	$\mathbf{G}_{\mathcal{A}}^{\text{unif}}(\lambda, w)$
sample B honestly (from the public seed)	same
$(\mathbf{r}_i, \mathbf{e}_{2,i})_{i=1}^w$ from the branch distributions	— — —
$\mathbf{y} := B \sum_{i=1}^w \mathbf{r}_i + \sum_{i=1}^w \mathbf{e}_{2,i}$	$\mathbf{y} \leftarrow \$ R_q^n$
$b' \leftarrow \mathcal{A}(B, \mathbf{y})$	$b' \leftarrow \mathcal{A}(B, \mathbf{y})$
return b'	return b'

Figure 5: The SDI games at width w . The structured game’s challenge is an honestly formed width- w branch sum under the public matrix; the uniform game’s is a random vector.

Four remarks delimit the assumption. First, SDI is single-instance decision module-LWE in normal form: B is the public matrix, the secret $\mathbf{R}$ and error $\mathbf{E}_2$ are short (width- w sums of bounded distributions), and there are n ring samples. No barrier paragraph is needed here, in contrast to Section 5.6: the hidden quantity is the honest user’s blinding, formed exactly as the construction prescribes, so nothing blocks a reduction to a standard decision-MLWE instance along known lines (bounded, high-entropy secret and error distributions are handled by the usual re-randomization and robustness arguments); what is genuinely needed is the concrete costing below, not a new idea. Second, the assumption is used at width 1 in the proof (Theorem 5.13) and at width κ in the tightness direction; both ends matter, and at the parameters of Section 6 they are attacked at similar cost. Third, the public matrix: B is seed-derived, hence uniform; grinding the seed gives the adversary a choice among at most Q_R uniform matrices, and the assumption is understood with this factor absorbed. Fourth, the quantum analogue is open, as for ssHMLWE.

Costing SDI. This is a lattice-estimator question, and a preliminary look says it deserves care. The generic decider is the *dual distinguishing attack*: a vector $(\mathbf{w}, B^T \mathbf{w} \bmod q)$ of length W in the rank- $2n$ module lattice $\{(\mathbf{w}, \mathbf{z}) : \mathbf{z} = B^T \mathbf{w} \bmod q\}$ (dimension $2nd$ over $\mathbb{Z}$, determinant q^{nd}) gives the statistic $\langle \mathbf{w}, \mathbf{y} \rangle = \langle B^T \mathbf{w}, \mathbf{R} \rangle + \langle \mathbf{w}, \mathbf{E}_2 \rangle \bmod q$, which for structured $\mathbf{y}$ is confined to roughly $\pm W\sqrt{w}(\beta_r + \beta_2)$ up to small constants (writing β_r, β_2 for the branch bounds) and is uniform for uniform $\mathbf{y}$, provided $\mathbf{w}$ has a component that is a unit in R_q . Such vectors exist

down to the Gaussian heuristic $W \approx q^{1/2} \sqrt{2nd/(2\pi e)}$, so a distinguishing vector exists whenever $\sqrt{w}(\beta_r + \beta_2) \sqrt{2nd/(2\pi e)} \lesssim q^{1/2}$ — a condition the parameters of Section 6 meet with a wide margin at every width $w \leq \kappa$, because the branch bounds are tiny next to $\sqrt{q}$. The attack’s cost is therefore not the existence of the statistic but the lattice reduction that finds a good enough vector, in dimension $2nd = 11,904$, with required quality set by $\log(q/(\sqrt{w}(\beta_r + \beta_2) \sqrt{nd}))$ rather than by κ ’s exponential lane floor. A preliminary sieve-cost estimate (Gaussian heuristic, blocksize in the mid hundreds; to be replaced by an LWE-estimator run [5]) puts the distinguishing attack on the order of 2^{160} classical and 2^{145} quantum operations. That is above the 128-bit target but *below* the $2^{\kappa/2}$ lane floor: at the current parameters the cheapest attack on the construction may be this privacy attack, not the splice. Exact costing, the module structure included, belongs to the parameter analysis (item 7 of Section 7).

The theorem. Write B_n for the worst-case per-coefficient size of the correctness error of Section 4 with the response error at its bound — of order $\kappa(\beta_s\beta_2 + \beta_e\beta_r) d$ — adversarial choice of the response error included.

Lemma 5.9 (key pinning). *Except with probability $\varepsilon_{\text{sound}} + \varepsilon_{\text{pok}}$, every session that $\mathcal{A}$ completes in $\text{Exp}^{\text{blind}}$ satisfies $h = \mathbf{s}^T \mathbf{C}_{\text{sel}} + e$ with $\|e\| \leq \beta_e$ for a key $(\mathbf{s}, \mathbf{e}_1)$ that the experiment holds before the credentials are delivered, with $\mathbf{t}^T = \mathbf{s}^T B + \mathbf{e}_1^T$, the norm bounds, and some component of $\mathbf{s}$ a unit in R_q .*

Proof. A verifying π_{resp} proves knowledge of $(\mathbf{s}, \mathbf{e}_1, e)$ satisfying $\mathcal{R}_{\text{resp}}$; the experiment runs the extractor on $\mathcal{A}$ ’s first verifying proof and (KV)-checks the result; failure of the proof, the extraction, or the checks aborts both sessions by convention, and a session that aborts contributes exactly $1/2$ to the win probability under (ABORT). Alternative valid representations of the same $\mathbf{t}$ differ by a short kernel element of B and are handled as in Theorem 5.5; the experiment fixes the extracted representation. $\square$

Lemma 5.10 (rounding erases the noise). *Condition on Lemma 5.9. Except with probability $\varepsilon_b \leq 8d B_n p/q$ over the user’s coins, every delivered credential satisfies*

$$v = \lceil \mathbf{s}^T \mathbf{U}^* \rceil_p,$$

independent of the response error and of the correctness noise; and the event that exactly one of the two delivered credentials verifies is contained in the same boundary event, so validity patterns carry no information about b outside ε_b .

Proof. By pinning, $v = \lceil h - \mathbf{t}^T \mathbf{R}^* \rceil_p = \lceil \mathbf{s}^T \mathbf{U}^* + \nu + e \rceil_p$ with $\nu = \mathbf{s}^T \mathbf{E}_2^* - \mathbf{e}_1^T \mathbf{R}^*$ (using $\mathbf{t}^T = \mathbf{s}^T B + \mathbf{e}_1^T$), and every coefficient of $\nu + e$ is at most B_n in the worst case, adversarial e at its bound included. Since the extracted $\mathbf{s}$ has a unit component and $\mathbf{U}^*$ is uniform, $\mathbf{s}^T \mathbf{U}^*$ is uniform in R_q ; it lands within B_n of a rounding boundary in a given coefficient with probability at most $2B_n p/q$. Outside that event the rounding is unchanged by $\nu + e$. Union over d coefficients and two credentials. The verification equation compares v with $\lceil \mathbf{s}^T \mathbf{U}^* \rceil_p$ exactly, so a credential fails to verify only inside the boundary event, and the event that exactly one of the two fails is a union of such events. $\square$

Lemma 5.10 is what makes a proof possible at all. The response $h = \mathbf{s}^T \mathbf{C}_{\text{sel}} + e$ is computed by the adversary and carries its per-session error e ; without rounding, v would fingerprint the session. The rounding window is enormous next to the noise (q/p against B_n per coefficient), so outside the boundary event the fingerprint is destroyed.

Lemma 5.11 (lane decoupling). *Fix a session and a lane i , and consider two adjacent hybrids. In $\mathbf{G}^{(i-1)}$, lane i 's selected branch is honest: seed-generated, leaf hashed, its vector inside the aggregate. In $\mathbf{G}^{(i)}$ it is virtual: the experiment samples $(m_i^*, \rho_i^*, \mathbf{u}_i^*)$ at random, programs $H_R(\text{credential-lane}, i, m_i^*, \rho_i^*) = \mathbf{u}_i^*$, sends a uniform ℓ_i , and replaces the branch's contribution to $\mathbf{C}_{\text{sel}}$ by an independent uniform vector. Then for every adversary $\mathcal{A}$ distinguishing the two hybrids with advantage δ there is an SDI(1) distinguisher $\mathcal{B}$ of essentially the same running time with $\text{Adv}_{\mathcal{B}}^{\text{SDI}(1)}(\lambda) \geq \delta - O(Q_R 2^{-256})$.*

Proof. $\mathcal{B}$ receives $(B, \mathbf{y}^*)$ and simulates the hybrid. It programs Expand_B on $\mathcal{A}$'s seed to output B (the seed-grinding union is absorbed into SDI, as remarked above); it generates every other lane of both sessions honestly, opened and selected alike, as the hybrid prescribes; for lane i it samples $(m_i^*, \rho_i^*, \mathbf{u}_i^*)$, programs H_R , computes $\ell_i = H_{\text{leaf}}(\text{sid}, i, b_i^*, \mathbf{y}^* + \mathbf{u}_i^*)$ by querying its own oracle in the structured case and samples ℓ_i uniformly in the uniform case; it sets the lane's contribution to $\mathbf{C}_{\text{sel}}$ to $\mathbf{y}^* + \mathbf{u}_i^*$, sets $\mathbf{c}_{\text{agg}} = \mathbf{C}_{\text{sel}} + \sum_i \mathbf{c}_{i, \chi_i}$, and programs H_{chal} on the assembled input to the challenge that opens the intended side of every lane. At delivery it answers with the credential $v = \lceil \mathbf{s}^T \mathbf{U}^* \rceil_p$ (Lemma 5.10), using the extracted key (Lemma 5.9). In the structured case this is exactly $\mathbf{G}^{(i-1)}$, in the uniform case exactly $\mathbf{G}^{(i)}$, unless one of two events occurs. (a) $\mathcal{A}$ queries a random oracle at a point the simulation programmed, before that point exists: each programmed input carries at least 256 fresh bits, so this costs $O(Q_R 2^{-256})$. (b) $\mathcal{A}$ queries G , H_ρ , or H_{leaf} at lane i 's true preimage material, which would let it verify lane i beyond what the hybrids prescribe: the seed, the tag ρ_i^* , and the leaf ℓ_i are each 256-bit random-oracle images of material $\mathcal{A}$ does not hold, and the aggregate is no help, since stripping lane i out of $\mathbf{C}_{\text{sel}}$ requires the other lanes' contributions — virtual ones are uniform and unknown, honest ones are seed-locked — so (b) is preimage finding against a 256-bit image and costs $O(Q_R 2^{-256})$. Outside (a) and (b), $\mathcal{A}$'s distinguishing advantage is $\mathcal{B}$'s. $\square$

Lemma 5.12 (ideal world). *With both sessions fully virtual (every lane decoupled), and conditioned on Lemmas 5.9 and 5.10, the adversary wins with probability exactly 1/2.*

Proof. Each session's view is now $W = (\text{sid}, \text{opened branches}, \{\ell_i\}, \mathbf{c}_{\text{agg}}, \chi, h)$ with the opened branches honest and independent, the ℓ_i uniform, and $\mathbf{C}_{\text{sel}} = \mathbf{y} + \mathbf{U}^*$ with $\mathbf{y}$ uniform and independent of everything else; $h = \mathbf{s}^T \mathbf{C}_{\text{sel}} + e$ is a function of $\mathbf{C}_{\text{sel}}$ and $\mathcal{A}$'s own short e , which $\mathcal{A}$ chose before the credential material (m_i^*, ρ_i^*) was fixed anywhere in its view. The credential is $C = ((m_i^*, \rho_i^*)_i, v = \lceil \mathbf{s}^T \mathbf{U}^* \rceil_p)$. Conditioned on C , the vector $\mathbf{C}_{\text{sel}} = \mathbf{y} + \mathbf{U}^*$ is uniform (over $\mathbf{y}$), so $(\mathbf{C}_{\text{sel}}, h)$ — and with it $\mathbf{c}_{\text{agg}}, \chi$, and the whole view — is independent of C . Each session therefore contributes a pair (W_j, C_j) with $W_j \perp C_j$, the two pairs are independent and identically distributed, and the adversary receives (W_0, W_1, C_b, C_{1-b}) . Swapping b exchanges the second components of two independent identically distributed pairs, which leaves the distribution unchanged; every strategy has $\Pr[b' = b] = 1/2$. If either session aborted, the delivery is $(\perp, \perp)$ and the same statement holds trivially. $\square$

Theorem 5.13 (dishonest-signer blindness, conditional). *Assume SDI at width 1 (Game 5); assume the response proof is knowledge-sound for $\mathcal{R}_{\text{resp}}$, with soundness and extraction errors $\varepsilon_{\text{sound}}$ and ε_{pok} ; and assume (KV), (SID), (ABORT). Then for every adversary $\mathcal{A}$ in $\mathbf{Exp}^{\text{blind}}$ making at most Q_R random-oracle queries there is a reduction $\mathcal{B}$, of essentially the same running time, with*

$$\text{Adv}_{\mathcal{A}}^{\text{blind}}(\lambda) \leq 2\kappa \text{Adv}_{\mathcal{B}}^{\text{SDI}(1)}(\lambda) + \varepsilon_{\text{sound}} + \varepsilon_{\text{pok}} + \varepsilon_b + O((\kappa Q_R + Q_R^2) 2^{-256}).$$

Proof. Chain the lemmas. Lemma 5.9 pins the key and the responses at cost $\varepsilon_{\text{sound}} + \varepsilon_{\text{pok}}$, with aborted sessions contributing exactly 1/2 under (ABORT). Lemma 5.10 replaces every delivered

credential by $v = \lceil \mathbf{s}^T \mathbf{U}^* \rceil_p$ at cost ε_b and disposes of validity-pattern strategies. Then apply Lemma 5.11 lane by lane across both sessions: 2κ hops, each costing $\text{Adv}^{\text{SDI}(1)} + O(Q_R 2^{-256})$, after which both sessions are fully virtual and Lemma 5.12 gives $\Pr[b' = b] = 1/2$. Collecting terms gives the bound; concurrency is no obstacle, since every hop’s reduction answers $\mathcal{A}$ ’s queries online. $\square$

Tightness: SDI is necessary, and it may bind below the lane floor. An $\text{SDI}(\kappa)$ decider breaks blindness outright: given the two sessions and the two delivered credentials, compute $\mathbf{V}_{jk} = \mathbf{C}_{\text{sel},j} - \mathbf{U}_k^*$ for $j, k \in \{0, 1\}$; the matched pair is structured, the crossed pair is uniform-masked, and deciding which is which is SDI at width κ . So blindness for this construction is sandwiched between SDI at width 1 (sufficient, Theorem 5.13) and SDI at width κ (necessary) — the two ends of one parametric assumption, attacked at similar cost by the dual distinguisher of the costing paragraph. In particular the leaf hashes of the selected branches, though visible to the signer, do not weaken unlinkability beyond SDI: they are one-way images of seed-locked material, and Lemma 5.11 prices every attempt to open them at 2^{-256} per query. What the proof cannot repair is the statistic itself: if the dual distinguishing attack of the costing paragraph is as cheap as estimated there, it links presentations to issuance sessions below the lane floor, and blindness — not the splice — is the binding constraint of the construction at the parameters of Section 6.

A statistical variant. The lane-decoupling hops are the only computational steps of the proof, and they become statistical if the branch error is wide: once a single branch’s $\mathbf{e}_2$ has width within a small factor of $\sqrt{q} / \sqrt{2nd}$, the distribution of $B\mathbf{r} + \mathbf{e}_2$ is within negligible statistical distance of uniform over R_q^n by the standard smoothing argument [6], Lemma 5.11 comes free at every lane, and blindness holds unconditionally in the random-oracle model. The price is correctness slack: the correctness error inherits the wider $\mathbf{e}_2$, several dozen bits per coefficient against the same rounding window, which the parameters of Section 6 do not have; restoring the same rounding-failure bound would mean growing q by roughly that amount (on the order of 30–40% more commitment bandwidth). Whether the trade is worth it belongs to the same parameter analysis.

What an implementation must enforce. The three conventions are load-bearing. Without (KV) the adversary plants a key whose B has a trapdoor, or whose $\mathbf{s}$ is a zero divisor in every component so that $\mathbf{s}^T \mathbf{U}^*$ is constant and the rounding erases nothing. Without (SID) the two users’ hash domains are no longer separated by their identifiers and the per-session programming in Lemma 5.11 is unjustified (the unforgeability side needs session uniqueness for much stronger reasons, Section 5.7). Without (ABORT), failing exactly one session on purpose — one malformed h — reveals b from the delivery pattern alone. The response proof must be a proof of knowledge, since Lemmas 5.10 and 5.11 compute with the extracted key; its zero-knowledge property is not needed here, because the proof covers values the adversary already holds, though it is needed on the unforgeability side.

5.10 What this construction gives up

Relative to BLNS23, the compactness of this design costs three things:

1. **The proof route.** The BLNS23 reduction to HMLWE exists because the user’s zero-knowledge proof hands the reduction the full witness of every request. Cut-and-choose removed that proof, and the response relation gives a reduction nothing to replace it with: it

cannot evaluate $\mathbf{s}^T \mathbf{C}_{\text{sel}}$ on any input that is not a fresh hash image, and the peeled sum generally is not one. Section 5.6 sharpens the obstacle to one term: with a single key, the junk contribution $\mathbf{s}^T \mathbf{w}$ of an unopened, never-checked branch is exactly what no HMLWE-style oracle can supply.

2. **Extraction handles.** The leaf hashes commit to every branch, but only the opened preimages are ever revealed: the selected branches remain hidden behind their hashes, so no forking or seed-tag technique can recover them. A proof would have to reintroduce extractable per-branch commitments with transmitted openings — giving up peeling and much of the compactness — or rely on the random-oracle analysis of Section 5.5, which covers the challenge game, together with the isolated assumption of Section 5.6 for the lattice step.
3. **Defense in depth.** With one key, every lane’s value lives under the same secret $\mathbf{s}$, so any lane vector can cancel any other: cross-lane cancellation is algebraically possible, and only its cost — imposed by the fixpoint wall — prevents it. The commit-before-challenge order is the single barrier; there is no second line of defense behind it.

6 Efficiency sketch

Use the BLNS23 keyed-verification values: $d = 64$, $n = 93$, $\lceil \log_2 q \rceil = 152$, $p = 4$, 32-byte messages and tags, and $\kappa = 512$ lanes as the post-quantum starting point. Ring elements are packed at 152 bits per coefficient, so one ring element occupies 1,216 bytes and one commitment vector occupies $93 \cdot 64 \cdot 152/8 = 113,088$ bytes.

The compactness comes from exactly two features of the design, and it is worth being explicit about both.

One aggregate commitment. Because every lane is evaluated under the same key, the signer needs only the *sum* of the selected vectors, and that sum can be recovered by subtraction: the user sends one commitment vector $\mathbf{c}_{\text{agg}}$ (113,088 bytes) and the signer peels. The unopened branch vectors are never transmitted at all. Binding the branch table costs κ additional hashes: the user sends the leaf hashes of the unopened branches (16,384 bytes) and the signer recomputes the opened half from the seeds. This is the price of the fix described in Section 3.2: without the leaf list the challenge would bind no branch table at all.

Revealed seeds instead of openings. An opened branch is fully determined by its 32-byte seed: the signer regenerates the message, the blinding vectors, the tag, the commitment vector, and the leaf hash locally and checks them. The κ opened branches therefore occupy 16,384 bytes instead of κ explicit vectors with witnesses.

Two further consequences are worth noting. The response proof covers a single-key relation — one inner product and one public-key equation — so it is small in principle, and unlike the BLNS23 user proof it contains no hash evaluations. And the signer’s long-term state is one secret vector and one matrix, regardless of κ .

What does *not* shrink: the credential itself. Presentation still carries κ tags and messages plus the rounded value v , since every lane contributes an independently hidden message. The savings are entirely in issuance bandwidth, signer state, and proof complexity.

Computation shifts as follows. The user expands $2\kappa = 1,024$ seeds and builds 1,024 branch vectors offline. The signer’s online work is κ seed expansions, κ formation checks, κ leaf-hash eval-

Wire component	Bytes
Aggregate commitment $\mathbf{c}_{\text{agg}}$	113,088
512 opened seeds (32 B each)	16,384
512 unopened leaf hashes (32 B each)	16,384
Session identifier	32
User $\rightarrow$ signer, fixed total	145,888
Aggregate response h	1,216
Signer $\rightarrow$ user	$1,216 + \pi_{\text{resp}} $
MAC $(\rho_0, \dots, \rho_{511}, v)$	16,400
MAC plus 512 32-byte messages	32,784

Table 1: Preliminary exact wire sizes in bytes at $\kappa = 512$. The first message is about 142.5 KiB, dominated by the single aggregate vector; the credential itself is about 32 KiB. The response proof is not instantiated, so its size is excluded.

uations, κ subtractions, one inner product, and one single-key proof. Issuance latency is therefore dominated by the seeded checks, not by the signing operation.

7 What a proof would still need

The random-oracle analysis of Section 5.5 proves the challenge-game content of the fixpoint property, and Section 5.6 reduces the remaining goals to one decisional assumption. A full security proof of the construction would additionally need:

1. **Extraction of selected branches.** The BLNS23 reduction to HMLWE exists because the user’s zero-knowledge proof hands the reduction the full witness of every request. Cut-and-choose removed that proof; peeling replaced per-branch vector transmissions by hash commitments whose selected preimages are never revealed, so partial extraction techniques (forking, seed tags) have nothing to work with. A reduction-based proof of this construction must either reintroduce individually committed branches with extractable openings — giving up the aggregate’s compactness — or avoid extraction entirely.
2. **A quantum random-oracle analysis.** The proof of Theorem 5.4 samples the challenge classically at query time and takes union bounds over classical query transcripts; a quantum attacker queries in superposition, and the analysis needs compressed-oracle or equivalent techniques. The quantum analogue of the structured-sum HMLWE assumption of Section 5.6 is open as well.
3. **Cryptanalysis or reduction of the lattice assumption.** Section 5.6 isolates the lattice step as structured-sum HMLWE and shows that no reduction from plain MLWE exists along known lines: the single key puts the committed junk term $\mathbf{s}^T \mathbf{w}$ out of reach of any HMLWE-style oracle. What remains is direct cryptanalysis of the assumption — the falsifiability criterion of Section 5.6 names what a break would look like — or a fundamentally different reduction.
4. **One-more simulation details.** The counting argument is now conditional (Theorem 5.8): with the common-message binding, an extra distinct message forces fresh lane commitments

in every lane, and the required value is a rounding of fresh random-function values. What remains are the hybrid pieces named in that theorem — response-proof simulation, key validation, and the correctness bound — instantiated for this construction.

5. **Sessions and grinding hygiene.** Session identifiers must never repeat within a key epoch, including under concurrency, rollback, and replication; the signer’s fresh value must be in place before the user computes the challenge.
6. **The blindness assumption.** Section 5.9 carries the dishonest-signer blindness proof through: every hop is proven except short-decomposition indistinguishability (Game 5), which is necessary as well as sufficient. What remains is a reduction of SDI to a standard decision-MLWE instance (robustness bookkeeping for the width- w sum distributions), its concrete costing against the dual distinguishing attack — at face value below the lane floor, and so possibly the binding constraint of the whole construction — its quantum analogue, and enforcement of the operational conventions (KV), (SID), (ABORT).
7. **Parameters.** A joint analysis of lattice hardness, the κ -term correctness error, rounding failure, the SDI distinguishing cost of Section 5.9 — the dual attack in dimension $2nd$, which at face value binds below the lane floor — and the $2^{\kappa/2} / 2^{\kappa/4}$ splice floor, together with the sum-to-target safety condition $N \gg \kappa (1 + \log_2 \kappa)$ of Section 5.4; $\kappa = 512$ is a starting point, not an analysis.

8 Conclusion

Cut-and-choose can replace the expensive well-formedness proof of BLNS23, and the aggregate-peeling form explored here is a compact instantiation: one signer key, one summed commitment, one leaf-hash list, one opened-seed set, and a first message of about 142.5 KiB at $\kappa = 512$ lanes. The commit-before-challenge order — over the aggregate and the branch table alike — is the single load-bearing element: it forces any attempt to aim the signed sum into a guess-and-check search whose cost we conjecture to be about $2^{\kappa/2}$, and it is all that stands between the scheme and constant-cost key recovery. For the challenge game this is now a theorem in the classical random-oracle model (Theorem 5.4), and its bounds are tight: the known attacks — the rectangle, blocked by session freshness, and the balanced splice at $2^{\kappa/2}$ classical or $2^{\kappa/4}$ quantum work, its yield demoted to credential duplication by the common-message binding — sit exactly where the bounds stop protecting the construction, while direct sum-to-target aiming is pushed far beyond reach by the size of the ring (Section 5.4). Past the challenge game, the lattice step is now a single isolated assumption rather than an unanalyzed heuristic: structured-sum HMLWE, with key recovery and one-more unforgeability reduced to it, its linear-attack surface reduced to module-SIS or to the proven bounds, and a precise account of why no standard reduction reaches it (Section 5.6). What remains open is the assumption itself, its quantum analogue, and the simulation details of the one-more hybrids. On the privacy side, blindness is now a theorem up to a second isolated assumption — short-decomposition indistinguishability, necessary as well as sufficient — and its best known attack, a dual distinguishing attack on the peeled sum, costs less than the lane floor at the current parameters: the privacy side, not the forgery side, may set the real floor, and costing it exactly is part of the remaining parameter work (Section 5.9). We present the construction as a target for analysis, not for deployment.

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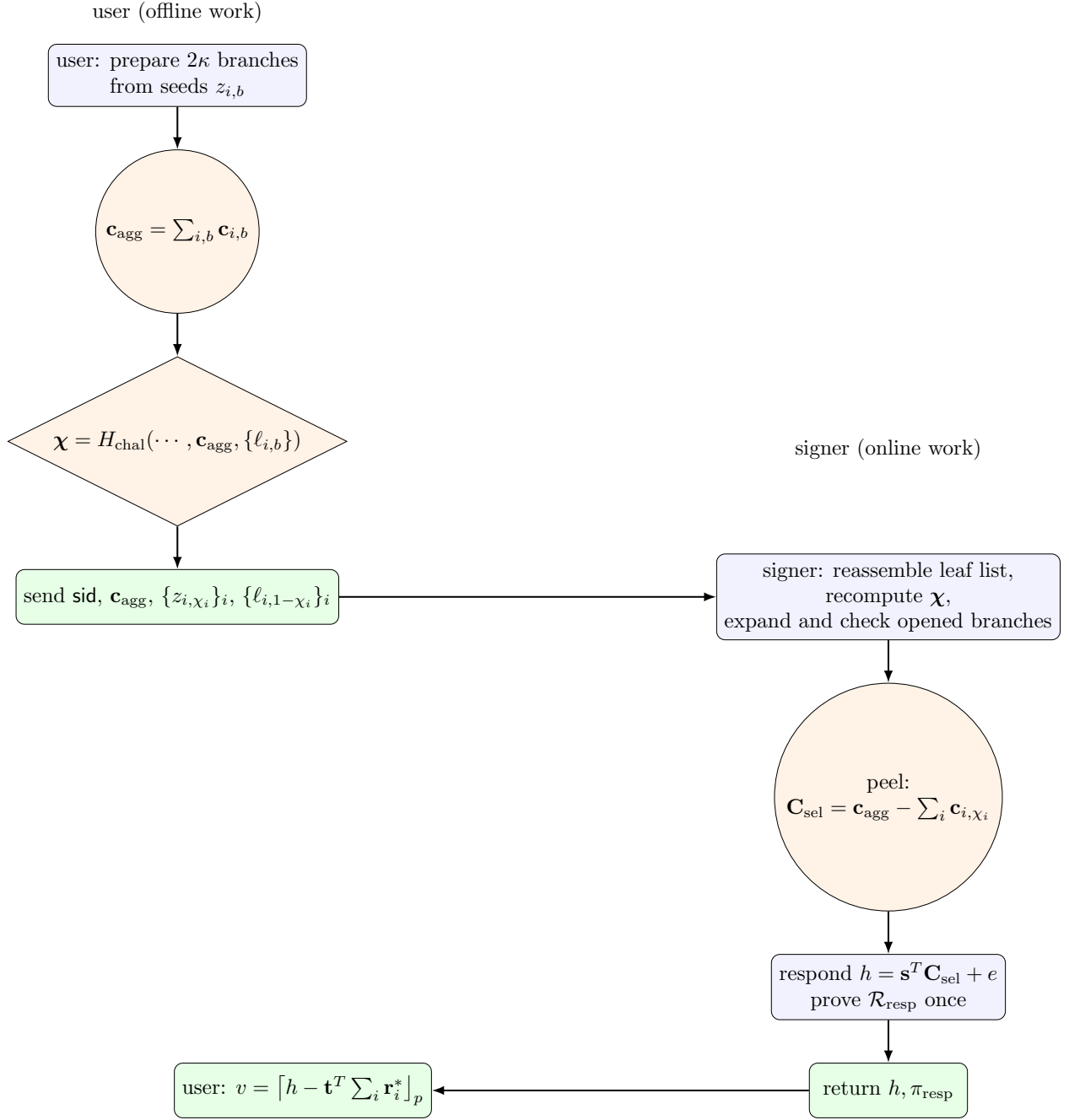


Figure 2: The two-message exchange. The user does all branch work offline and sends one aggregate vector, κ seeds, and κ leaf hashes. The signer’s online work is: reassemble the leaf list and recompute the challenge, expand and check κ opened branches, one subtraction per opened branch, one inner product under its single key, and one single-key proof.

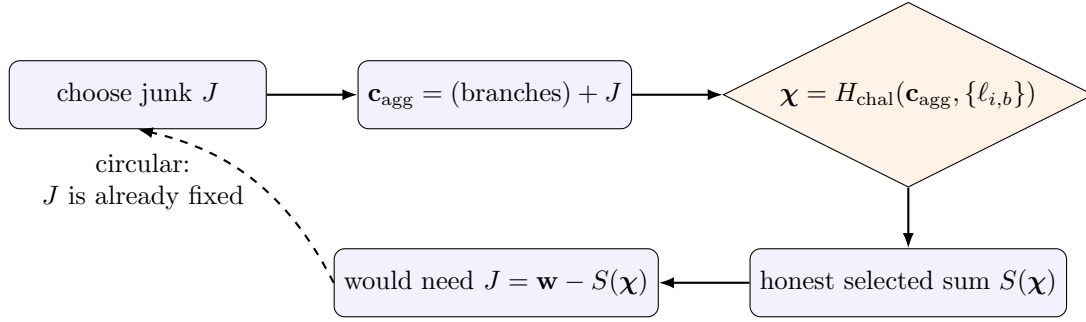


Figure 3: The fixpoint wall. Aiming the peeled sum at a target $\mathbf{w}$ requires junk that cancels a challenge-dependent honest sum; the junk is committed before the challenge exists. Resolving the cycle is a search: guess the challenge pattern, set the junk accordingly, and hope the hash returns the guessed pattern.